1.
On your Windows laptop or PC, open 'System Information' and take a screenshot showing your OS version, processor, and RAM details.

Checking System Information on a Windows Laptop or PC

Aim: To open System Information in Windows and identify the operating system version, processor, and installed RAM.

Step-by-Step Procedure

Step 1: Open System Information

1. Press **Windows + R** on your keyboard.
2. The Run dialog box will appear.
3. Type `msinfo32` in the Open field.
4. Press **Enter**.

Step 2: View System Summary

The System Information window will open. Select **System Summary** from the left panel.

Step 3: Identify the Required Details

Find the following information in the right panel:

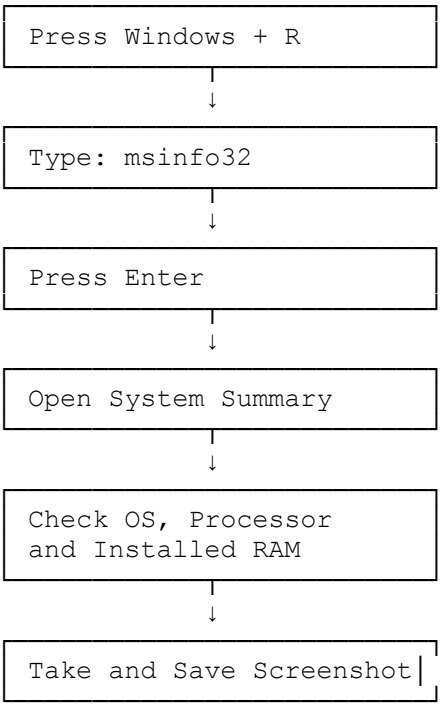
S.No.	Information	Description
1	OS Name and Version	Displays the installed Windows operating system and its version.
2	Processor	Displays the processor name and model installed in the computer.
3	Installed Physical Memory (RAM)	Displays the total amount of RAM installed in the computer.

Step 4: Take a Screenshot

1. Keep the System Summary window open.
2. Ensure that the OS Name, Version, Processor, and Installed Physical Memory (RAM) fields are visible.
3. Press **Windows + Shift + S** to open the Snipping Tool.
4. Select the rectangular snip option.
5. Drag over the System Information window to capture the required details.
6. Save the screenshot using the Snipping Tool.

Suggested screenshot filename: `System_Information.png`

Diagram: Steps to Open System Information



Observation Table

After opening System Information, fill in the values displayed on your computer.

Parameter	Your Computer's Details
Operating System	Write the OS name shown in System Information.
OS Version	Write the version shown in System Information.
Processor	Write the processor name shown in System Information.
Installed RAM	Write the installed physical memory shown in System Information.

Result

System Information was opened successfully, and the operating system version, processor, and installed RAM were identified. A screenshot should be attached to the assignment as evidence.

2.

Download any official Linux distribution ISO (like Ubuntu or Fedora), and use Rufus or BalenaEtcher to create a bootable USB drive for installation.

Hint: If you don't have a spare USB, use a virtual machine tool like VirtualBox to mount the ISO and start the installation process virtually.

Downloading a Linux ISO and Creating a Bootable USB Drive

Aim: To download an official Linux distribution ISO file and create a bootable USB drive using Rufus so that Linux can be installed on a computer.

Requirements:

- Windows laptop or PC
- USB flash drive (8 GB or larger recommended)
- Internet connection
- Rufus software
- Official Linux ISO file

Step-by-Step Procedure

Step 1: Download a Linux Distribution

1. Open a web browser.
2. Visit the official Ubuntu website: Ubuntu Desktop Download.
3. Download the Ubuntu Desktop ISO file.
4. Wait for the download to finish.
5. Locate the downloaded `.iso` file in the Downloads folder.

Example filename: `ubuntu-desktop-amd64.iso`

Step 2: Download Rufus

1. Visit the official Rufus website: Rufus Download.
2. Download Rufus for Windows.
3. Open the downloaded Rufus application.

Rufus is a tool that creates bootable USB drives from ISO files.

Step 3: Insert the USB Drive

1. Insert an empty USB flash drive into the computer.
2. Open Rufus.
3. Check the **Device** field.
4. Select the correct USB drive.

Important: Creating a bootable USB drive erases the existing data on that drive. Back up any important files before proceeding.

Step 4: Select the Linux ISO File

1. In Rufus, locate the **Boot selection** section.
2. Select **Disk or ISO image**.
3. Click **SELECT**.
4. Browse to the downloaded Ubuntu ISO file.
5. Select the file and click **Open**.

Step 5: Configure the Bootable USB

1. Check that the correct USB drive is selected.
2. Allow Rufus to choose the recommended settings automatically.
3. Check the partition scheme and target system settings for compatibility with the computer.
4. Click **START**.
5. If prompted to choose an image-writing mode, follow Rufus's recommended option.

Step 6: Confirm and Create the Bootable USB

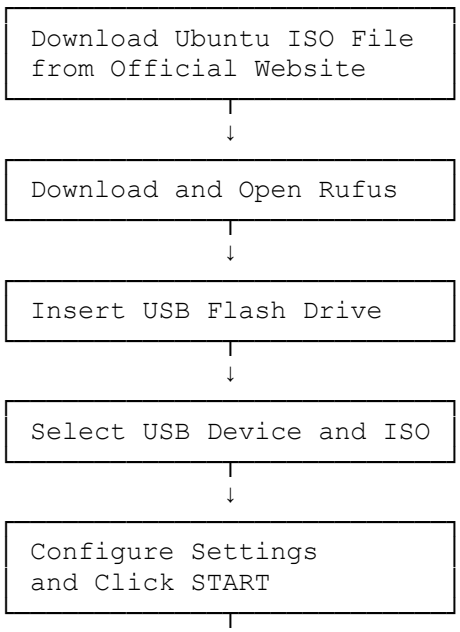
1. Read the warning that the USB drive will be erased.
2. Confirm only after checking that the correct USB drive is selected and important data is backed up.
3. Wait for Rufus to copy the Linux installation files.
4. When the status shows **READY**, the bootable USB drive is complete.
5. Close Rufus and safely eject the USB drive.

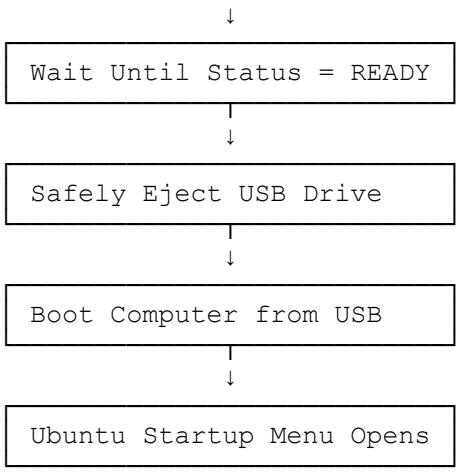
Step 7: Test the Bootable USB

1. Restart the computer with the USB drive inserted.
2. Open the computer's boot menu using the appropriate key for the device, such as F12, Esc, or F9.
3. Select the USB drive.
4. If the USB boots successfully, the Ubuntu startup menu should appear.
5. Select **Try Ubuntu** if available to test Linux without installing it.

Note: Do not select the installation option unless you intend to install Linux. Installing an operating system can modify or erase existing data, depending on the options selected.

Diagram: Creating a Bootable Linux USB





Alternative Method: Using VirtualBox

If you do not have a spare USB drive, you can use a virtual machine.

1. Download the Ubuntu ISO file.
2. Download and install Oracle VirtualBox.
3. Open VirtualBox and click **New**.
4. Enter a name, such as `Ubuntu`.
5. Select the downloaded ISO file as the installation image.
6. Allocate suitable RAM and virtual disk space.
7. Start the virtual machine.
8. Follow the Ubuntu installation or live-session instructions displayed on the screen.

A virtual machine allows you to run Linux inside Windows without creating a physical bootable USB drive.

Expected Result

S.No.	Task	Expected Result
1	Download the Ubuntu ISO	The ISO file is saved on the computer.
2	Open Rufus	Rufus starts successfully.
3	Select the USB and ISO	The correct device and image are selected.
4	Create the bootable USB	Rufus displays the READY status.
5	Boot from USB	The Ubuntu startup menu appears.

3.

On a macOS device (or using a macOS virtual machine), open 'System Preferences' and change the desktop wallpaper, then adjust the system language to Hindi or Gujarati.

Hint: Find 'Language & Region' in System Preferences to add or change languages.

. Changing Desktop Wallpaper and System Language in macOS

Aim: To change the desktop wallpaper and modify the system language to Hindi or Gujarati on a macOS device.

Requirements:

- A Mac computer or macOS virtual machine
- Access to System Preferences or System Settings

Part A: Changing the Desktop Wallpaper

Step 1: Open System Preferences

1. Turn on the Mac computer.
2. Click the **Apple menu (□)** in the top-left corner.
3. Select **System Preferences**.

Note: In newer versions of macOS, this option is called System Settings.

Step 2: Open Desktop & Screen Saver

1. In System Preferences, click **Desktop & Screen Saver**.
2. The desktop wallpaper settings will appear.

Step 3: Select a New Wallpaper

1. Browse the available desktop pictures.
2. Select your preferred wallpaper.
3. The desktop background will change automatically.

Step 4: Verify the Wallpaper

Return to the desktop and confirm that the selected wallpaper is displayed correctly.

Part B: Changing the System Language to Hindi or Gujarati

Step 1: Open Language & Region

1. Return to **System Preferences**.
2. Click **Language & Region**.
3. Find the Preferred Languages section.

Step 2: Add a Language

1. Click the **+ (Add)** button.
2. Search for **Hindi** or **Gujarati**.
3. Select your preferred language.
4. Click **Add**.

Step 3: Set the Preferred Language

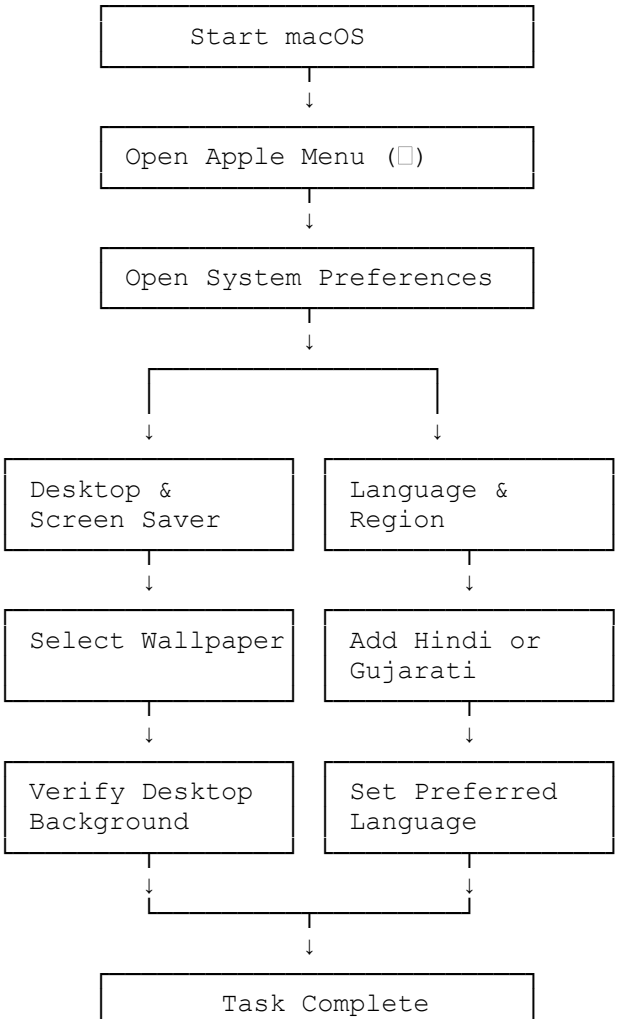
1. If prompted, choose whether to use Hindi or Gujarati as the primary language.
2. Confirm your selection.
3. If macOS asks you to restart or log out, follow the instructions to apply the change.

Step 4: Verify the Language

After the changes take effect, check the system menus and interface to see whether the selected language is displayed.

Note: The exact options vary by macOS version. On some versions, adding a preferred language does not automatically translate every application or system component.

Diagram: Changing Wallpaper and Language in macOS



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Expected Results

S.No.	Task	Expected Result
1	Open System Preferences	The settings window opens.
2	Change desktop wallpaper	The selected wallpaper appears on the desktop.
3	Open Language & Region	Language settings are displayed.
4	Add Hindi or Gujarati	The selected language appears in the preferred languages list.
5	Apply the language change	The system uses the selected language where supported.

4.
Install a fresh copy of Ubuntu Linux (real or virtual machine) and create a new user account named 'gamer'. After installation, enable automatic login for this user and take a screenshot of the login screen showing the username.

Install Ubuntu Linux and Create a User Account Named “gamer”

Aim

To install a fresh copy of Ubuntu Linux, create a new user account named **gamer**, enable automatic login for that user, and capture a screenshot showing the username.

Requirements

1. A computer or virtual machine.
2. Ubuntu Linux ISO file.
3. VirtualBox or a bootable USB drive.
4. Sufficient disk space and memory.

Procedure

Step 1: Download Ubuntu Linux

1. Open the official Ubuntu website.
2. Download the Ubuntu Desktop ISO file.
3. If using a virtual machine, install VirtualBox and create a new virtual machine.
4. Select the downloaded Ubuntu ISO file as the installation media.

Official website: Download Ubuntu

Step 2: Install Ubuntu

1. Start the computer or virtual machine using the Ubuntu ISO.
2. Select **Install Ubuntu**.
3. Choose the preferred language and keyboard layout.
4. Select the installation type and follow the on-screen instructions.
5. When prompted to create a user, enter **gamer** as the username.
6. Set a password and complete the installation.
7. Restart the system when installation finishes and remove the installation media if prompted.

Note: If Ubuntu is already installed, you can create the account in Step 3 without reinstalling the operating system.

Step 3: Create the "gamer" User Account

If you did not create the account during installation:

1. Open **Settings**.
2. Select **Users**.
3. Click **Unlock** and enter the administrator password.
4. Select **Add User**.
5. Enter the account details:
    - **Full Name:** Gamer
    - **Username:** gamer
    - **Account Type:** Standard
6. Set a password and click **Add**.

The new account named `gamer` will now be available.

Step 4: Enable Automatic Login

1. Open **Settings**.
2. Navigate to **System → Users** or **Users**, depending on the Ubuntu version.
3. Click **Unlock** and enter the administrator password.
4. Select the **gamer** account.
5. Enable **Automatic Login**.
6. Close Settings.

Important: Automatic login allows the system to open the selected account without asking for its password at startup. Use this setting only on a computer where unauthorized physical access is not a concern.

Step 5: Restart and Verify

1. Restart Ubuntu.
2. Check whether the system opens the `gamer` account automatically.
3. If the login screen appears instead, verify that automatic login is enabled for the correct account.

4. Open the terminal and run the following command to verify the current username:

```
whoami
```

Expected output:

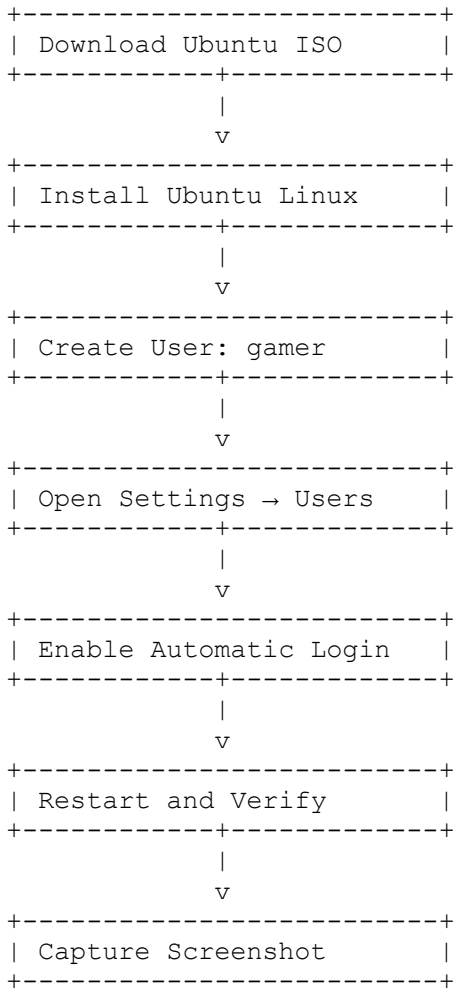
```
gamer
```

Step 6: Take a Screenshot

1. Capture the login screen if it displays the username `gamer`.
2. On Ubuntu, press **PrtSc** or open the Screenshot application.
3. Save the screenshot as `ubuntu_gamer_login.png`.

Important: With automatic login enabled, the login screen may not appear during startup. If your assignment specifically requires a screenshot of the login screen, you can temporarily disable automatic login, sign out, capture the screen showing `gamer`, and then re-enable automatic login. Alternatively, capture the Users settings showing that automatic login is enabled for `gamer`.

Diagram: Ubuntu Installation and Configuration



Expected Result

Item	Expected Result
Operating System	Ubuntu Linux
Username	`gamer`
User Account	Created successfully
Automatic Login	Enabled for `gamer`
Verification	`whoami` displays `gamer`
Screenshot	Saved as `ubuntu_gamer_login.png`

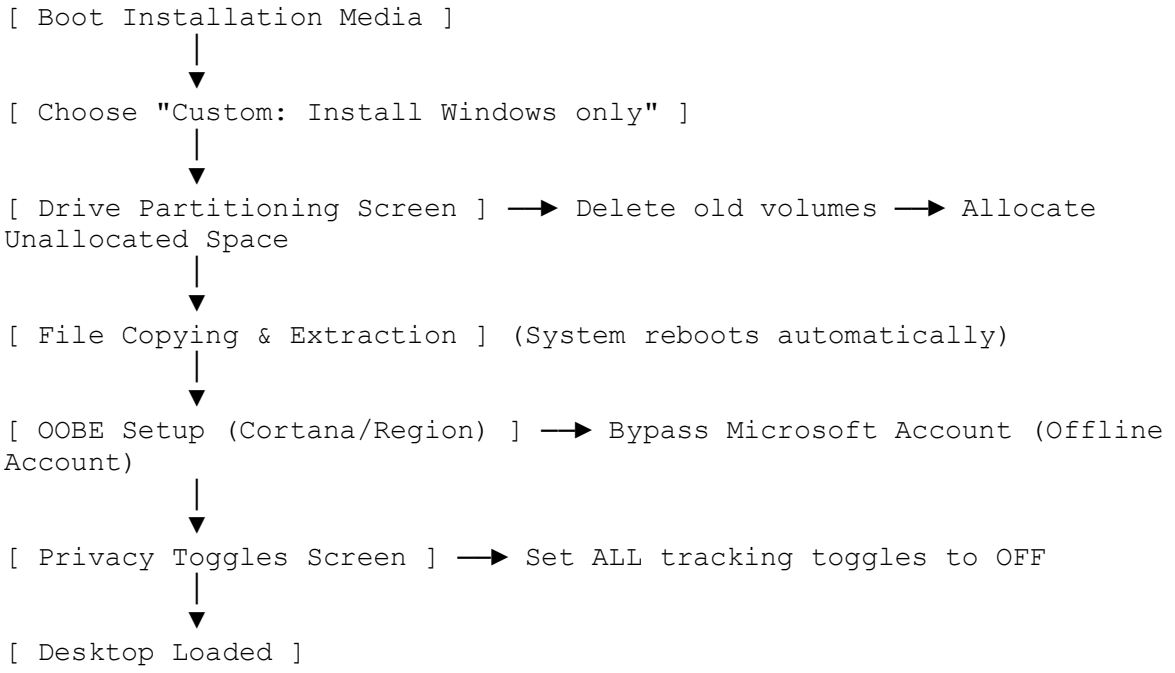
5.

Use ChatGPT to generate a step-by-step guide for installing Windows 10 on a laptop, including partitioning tips and post-installation settings for privacy and updates. Follow the guide and note any steps that were unclear or missing.

Comprehensive Guide: Installing Windows 10 from Scratch

Here is a comprehensive, step-by-step technical guide to installing a clean copy of Windows 10 on a laptop, complete with optimal drive partitioning practices and post-installation optimization.

Phase 1: Installation Workflow



Phase 2: Step-by-Step Installation

1. Boot from the Installation Media

- Insert your bootable Windows 10 USB flash drive into the laptop.
- Power on the machine and immediately spam the **Boot Menu Key** (typically `F12`, `F11`, `F9`, or `Esc` depending on your laptop brand).
- Select your **USB Drive** from the boot list.
- When prompted, press any key to boot into the Windows Setup wizard.

2. Initial Setup Configurations

- Select your Language, Time/Currency format, and Keyboard layout. Click **Next**, then click **Install Now**.
- Enter your 25-character product key. If you are reinstalling Windows on a machine that was previously activated, click **"I don't have a product key"**—Windows will automatically reactivate via digital license once connected to the internet.
- Select the correct edition of Windows 10 (Home or Pro) matching your license. Accept the software terms.

3. Drive Partitioning (Crucial Step)

- When asked *"Which type of installation do you want?"*, choose **Custom: Install Windows only (advanced)**.
- **The Clean Slate Strategy:** You will see a list of existing drives and partitions. Select each partition on your primary drive and click **Delete** until you are left with a single entry: `Drive 0 Unallocated Space`.
- **Partitioning Options:**
    - **Option A (Simple & Recommended for SSDs):** Select the unallocated space and click **Next**. Windows will automatically create the necessary hidden recovery/system partitions and format the remaining space as your `C:` drive.
    - **Option B (Data Separation):** If you want a separate partition for files/games, click **New**. Enter a smaller size for your operating system (e.g., `102400 MB` for a 100GB OS drive). Click **Apply**. Select the remaining unallocated space, click **New**, and click **Apply** to make your storage partition. Select the first primary partition you created and click **Next**.

4. The Out-of-Box Experience (OOBE)

- The laptop will copy files and restart. **Remove the USB drive** during the restart to prevent the laptop from looping back into the installation wizard.
- Select your Region and Keyboard layout.

Phase 3: Critical Post-Installation Settings

1. Privacy Toggles (During & After Setup)

- **During Setup:** Windows will present a screen titled *"Choose privacy settings for your device"*. **Toggle every option to OFF** (Diagnostic data to Basic, Inking & Typing to Off, Tailored experiences to Off, Advertising ID to Off, Location to Off).
- **After Setup:** Once on the desktop, go to **Settings (Win + I) > Privacy**.
    - Under **General**, uncheck all boxes.
    - Under **Background apps**, turn the master toggle **OFF** to stop unwanted apps from running down your laptop battery in the background.

2. Account Configuration

- To avoid being forced into using a Microsoft cloud account, disconnect your laptop from Wi-Fi during the initial setup phase.
- Choose **"I don't have internet"** followed by **"Continue with limited setup"**. This forces Windows to let you create a secure, offline **Local Account**.

3. Windows Update & Driver Optimization

- Reconnect to the internet once on the desktop.
- Go to **Settings > Update & Security > Windows Update** and click **Check for updates**. Run this multiple times until no updates remain. This installs essential hardware drivers (Wi-Fi, Bluetooth, trackpad, and graphics).
- **Pause updates capability:** If you want to prevent unexpected reboots during work, click **Advanced options** under Windows Update and configure your "Active hours" or pause feature updates.

Field Notes: Commonly Confused or Missing Steps

When executing this process, tech users frequently encounter these easily missed issues:

- **Missing Wi-Fi Drivers at Setup:** Newer laptops (especially those with Intel 12th Gen/AMD Ryzen 6000 series or newer) often do not have native Wi-Fi drivers on the Windows 10 installation media. You will need a secondary device to download the wireless driver from the manufacturer's site onto a USB, or use an ethernet adapter to pass the setup screen.
- **Drive Not Found (Intel VMD/RAID):** If the partitioning screen appears completely blank, your laptop likely has Intel Volume Management Device enabled. You must download the Intel Rapid Storage Technology (IRST) driver, extract it to your installation USB, click **Load Driver** on the partitioning screen, and select that folder to make your storage drive appear.
